

Assignment 1

1(i)

$$(10110010101)_{25}$$

Hint! the number is negative
since sign bit is 1

Now, performing 1's complement

$$(01001101010)_{1's}$$

+1

$$-(01001101011)_2$$

$$\rightarrow -(1 \times 2^0 + 1 \times 2^1 + 1 \times 2^3 + 1 \times 2^5 + 1 \times 2^6 + 1 \times 2^9)$$

$$\rightarrow -(619)_{10}$$

$$\therefore -(619)_{10} = -(1543)_7 \quad (\text{Ans!})$$

$$\begin{array}{r|l} 7 & 619 \\ \hline 7 & 88 \rightarrow 3 \\ 7 & 12 \rightarrow 4 \\ 7 & 1 \rightarrow 5 \\ & 0 \rightarrow 1 \end{array} \quad \begin{array}{c} \uparrow \text{LSB} \\ \uparrow \text{MSB} \end{array}$$

1(ii)

$$(00110010101)_{15}$$

$$\rightarrow (00110010101)_2$$

$$\rightarrow 1 \times 2^0 + 1 \times 2^2 + 1 \times 2^4 + 1 \times 2^7 + 1 \times 2^8$$

$$\rightarrow (405)_{10}$$

$$\begin{array}{r|l} 9 & 405 \\ \hline 9 & 45 \rightarrow 0 \\ 9 & 5 \rightarrow 0 \\ & 0 \rightarrow 5 \end{array} \quad \begin{array}{c} \uparrow \text{LSB} \\ \uparrow \text{MSB} \end{array}$$

$$\therefore (405)_{10} = (500)_9 \quad (\text{Ans!})$$

1(iii)

$$(0011)_{\text{BCD}}$$

$$(3)_{10}$$

Now,

$$\begin{array}{r|l} 5 & 3 \\ \hline & 0 \rightarrow 3 \end{array} \quad \begin{array}{c} \uparrow \text{LSB} \\ \uparrow \text{MSB} \end{array}$$

$$(3)_{10} = (3)_5 \quad (\text{Ans!})$$

1 (iv)

$$(01001010101111)_2$$

Here, the number is positive because the sign bit is 0.

$$\rightarrow (01001010101111)_2$$

$$\rightarrow (01001010101111)_2$$

$$\rightarrow (2^0 + 2^1 + 2^2 + 2^3 + 2^4 + 2^6 + 2^8 + 2^{10} + 2^{13})_{10}$$

$$\rightarrow (9567)_{10}$$

1 (vii)

$$(101010101000011)_2$$

Here, the sign bit is 1. So, the number is negative.

$$\therefore - (01010101011100)_2$$

$$\rightarrow - (1 \times 2^2 + 1 \times 2^3 + 1 \times 2^4 + 1 \times 2^5 + 1 \times 2^7 + 1 \times 2^9 + 1 \times 2^{11} + 1 \times 2^{13})$$

$$\rightarrow - (10940)_{10}$$

1vi)
 $(01001010101111)_2$

Here, the number is positive because the sign bit is 0.

$$\rightarrow (01001010101111)_2$$

$$\rightarrow (1 \times 2^0 + 1 \times 2^1 + 1 \times 2^2 + 1 \times 2^3 + 1 \times 2^4 + 1 \times 2^5 + 1 \times 2^6 + 1 \times 2^7 + 1 \times 2^{10} + 1 \times 2^{13})$$

$$\rightarrow (9567)_{10}$$

1v)
 $(101010101000011)_2$

Here, the sign bit is 1. So, the number is negative.

$$\therefore (010101010111100)_{10}$$

+ 1

$$\therefore - (010101010111101)_2$$

next, $- (1 \times 2^0 + 1 \times 2^2 + 1 \times 2^3 + 1 \times 2^4 + 1 \times 2^5 + 1 \times 2^7 + 1 \times 2^9 + 1 \times 2^{11} + 1 \times 2^{13})_{10}$

$$= - (10941)_{10}$$

1viii)
 $(1100011)_{10}$

Here, each digit is in decimal. Now adding 4 to each digit,

$$\rightarrow (5544455)_{10}$$

$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 001 001 000 000 000 001 001

$$\rightarrow (010101010100010001010101)_{10} \text{ Exceeds 4}$$

$$\begin{array}{r}
 \textcircled{2} \textcircled{a} \textcircled{b} \ 1245 \) \ 435342125 \ (\ 310114 \\
 \underline{4223} \\
 1304 \\
 \underline{1245} \\
 1521 \\
 \underline{1245} \\
 2322 \\
 \underline{1245} \\
 10335 \\
 \underline{5512} \\
 423
 \end{array}$$

$$\begin{array}{l}
 1245 \times 1 = 1245 \\
 1245 \times 1 = 2534 \\
 1245 \times 3 = 4223 \\
 1245 \times 4 = 5512 \\
 1245 \times 5 = 11201
 \end{array}$$

$$\begin{array}{r}
 \textcircled{4} \ (412)_6 \\
 + \ (134)_6 \\
 \hline
 (550)_6
 \end{array}$$

$$4 + 2 = 6 \div 6 = 0 (s)$$

$$6 \div 6 = 1 (e)$$

$$3 + 1 + \overset{1}{1} = 5 (s)$$

Base-6

$$\begin{array}{r}
 412 \\
 - 134 \\
 \hline
 (234)_6
 \end{array}$$

$$\therefore (234)_6 = 5 \times 6^2 + 5 \times 6^1 = (210)_{10}$$

$$\therefore (234)_6 = 2 \times 6^2 + 3 \times 6^1 + 4 \times 6^0 = (94)_{10}$$

Base-6

$$\begin{array}{r}
 412 \\
 \times 134 \\
 \hline
 2452 \\
 2040x \\
 412xx \\
 \hline
 (104432)_6
 \end{array}$$

$$\begin{aligned}
 \therefore (104432)_6 &= 1 \times 6^5 + 4 \times 6^4 + 4 \times 6^3 + 5 \times 6^2 + 2 \times 6^1 \\
 &= (8816)_{10}
 \end{aligned}$$

Now,

$$(412)_6 = 4 \times 6^2 + 1 \times 6^1 + 2 \times 6^0 = (152)_{10}$$

$$(134)_6 = 1 \times 6^2 + 3 \times 6^1 + 4 \times 6^0 = (58)_{10}$$

Here, Base-10

$$\begin{array}{l}
 4002134 \rightarrow \\
 152 - 58 \rightarrow (94)_{10} \\
 152 + 58 \rightarrow (210)_{10}
 \end{array}$$

$$152 \times 58 = (8816)_{10}$$

$$(3) \quad (-5) - 6 \Rightarrow (-5) + (-6)$$

4 bit system

$$+6 \rightarrow (0110)_2$$

$$+5 \rightarrow (0101)_2$$

$$-6 \rightarrow (1001)_{1s}$$

$$-5 \rightarrow (1010)_{1s}$$

$$-6 \rightarrow \begin{array}{r} \\ (1001)_{1s} \\ +1 \\ \hline (1010)_{2s} \end{array}$$

$$-5 \rightarrow \begin{array}{r} \\ (1010)_{1s} \\ +1 \\ \hline (1011)_{2s} \end{array}$$

Now ,

$$-5 \rightarrow (1011)_{2s}$$

$$(-) \quad -6 \rightarrow (1010)_{2s}$$

$$\hline (10101)_{2s}$$

Carry out
(ignored/
removed)

indicates sign
0 $\rightarrow$ positive

$$\therefore (0101)_{2s}$$

There is overflow . Because addition of two negative numbers should give us a negative number. But, we've received a positive number as answer.

⑤ # 7 bit system in sign & magnitude representation.

$$\begin{aligned}
 \text{Smallest number} &= -(2^{n-1} - 1) \\
 &= -(2^{7-1} - 1) \\
 &\Rightarrow -(2^6 - 1) \\
 &\Rightarrow -(63)_{10} \\
 \therefore &-(3F)
 \end{aligned}$$

$$\# \boxed{n=7}$$

$$\begin{array}{rcl}
 16 & | & 63 \\
 \hline
 16 & | & 3 \rightarrow 15(F) \\
 \hline
 & & 0 \rightarrow 3
 \end{array}$$

$\uparrow$ LSB
 $\downarrow$ MSB

⑥ # 11 bit system in 1's complement.

$$\begin{aligned}
 \text{Largest number} &= +(2^{n-1} - 1) \\
 &= +(2^{11-1} - 1) \\
 &= (2^{10} - 1) \\
 &= (1023)_{10} \\
 \therefore &+(1777)_8
 \end{aligned}$$

$$\begin{array}{rcl}
 8 & | & 1023 \\
 \hline
 8 & | & 127 \rightarrow 7 \\
 \hline
 8 & | & 15 \rightarrow 7 \\
 \hline
 8 & | & 1 \rightarrow 7 \\
 \hline
 & & 0 \rightarrow 1
 \end{array}$$

$\uparrow$ LSB
 $\downarrow$ MSB

$$\textcircled{7} \text{ RAM, } (1C2)_{16} \rightarrow (2 \times 16^0 + 12 \times 16^1 + 1 \times 16^2)_{10} \\ = (450)_{10}$$

$$\text{Graphics card} = (10010110000)_2 \\ = (2^{10} + 2^7 + 2^5 + 2^4)_{10} \\ = (1200)_{10}$$

$$\text{Total cost} = (1200 + 2 \times 450)_{10} \\ = (2100)_{10}$$

$$\text{Money borrowed, } (4064)_8 = 4 \times 8^0 + 6 \times 8^1 + 4 \times 8^2 = (2100)_{10}$$

$$\text{Money remaining} = (2100)_{10} - (2100)_{10} \\ = (0)_{10} \quad \underline{\text{Ans}}$$